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Disentangling the Impacts of Changes in Climate and Vegetation on Hydrological Processes Across 2,252 Global Catchments

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Abstract Understanding large-scale hydrological responses to climate and vegetation change is critical, yet the propagation effects through the hydrological process on streamflow signatures in large scale are not quantified yet. Here, we analyzed 2,252 unregulated catchments worldwide to disentangle climate and vegetation change effects by employing a well-performed HBV-PML hydrological model, which features enhanced representations of evapotranspiration processes. Our analysis reveals consistent propagation patterns across most climate zones: (a) Increasing Leaf Area Index (LAI) enhances interception and transpiration but suppresses soil evaporation, decreases streamflow magnitude and dynamics, shortens high-flow durations, and increases low-flow frequency/duration, indicating reduced water availability, and vice versa. (b) The effects of precipitation changes are opposite to those of LAI. (c) Temperature modulates hydrological processes via rain-snow partitioning, soil evaporation, and evaporation from canopy interception. The impacts of climate and vegetation change are magnified as they propagate through total actual evapotranspiration to the remaining hydrological processes. Notably, the relative changes of key hydrological variables grow with increasing temporal scales. Contribution analysis identifies precipitation as the dominant driver of changes in streamflow signatures (73.6 ± 10.5% of catchments). However, contributions vary by climate zones and streamflow signatures, with median precipitation contributions exceed 60% in equatorial zones, while temperature dominates (61.4%–73.6%) low-flow changes in polar zones. This study provides global insights into hydrological responses under environmental changes and offers an essential process-based understanding of impact propagation mechanisms for water resources management.

Key Points:

- The HBV-PML performs well on streamflow simulation, reaching 0.65 of median Kling-Gupta Efficiency in the overall performance of streamflow signatures
- Hydrological propagation amplified climate and vegetation impacts, causing compounded streamflow changes over longer durations
- Precipitation dominates changes in streamflow signatures in 73.6 ± 10.5% of catchments, with over 60% contribution in equatorial regions

Supporting Information:

Supporting Information may be found in the online version of this article.

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1. Introduction

Climate and vegetation change are key drivers of hydrological processes (Gudmundsson et al., 2021; N. Wang et al., 2026). Amidst global climate change, observing variations in unregulated catchments provides a multi-dimensional perspective on hydrological responses to the internal changes in climate and vegetation, serving as a crucial baseline to further quantify human influences on hydrological processes. However, such quantification is not sufficient to disentangle hydrological regimes globally.

With the development of global climate and hydrological data sets, the impacts of climate and vegetation changes on hydrological processes have been extensively investigated (Abbas et al., 2022; Tan et al., 2020; S. Zhang et al., 2018). From the perspective of a global view, precipitation change contributes more than 70% to streamflow change in 1,116 global unimpacted catchments from 1981 to 2020 (H. Wei et al., 2024). In addition, the precipitation change is consistent with natural streamflow change across over 58% of global land areas (Huang et al., 2026). Elevated CO_2 shows important controls on streamflow changes through physiological alterations, such as stomatal closure, which can produce streamflow changes comparable in magnitude to those driven by precipitation in certain regions (Fowler et al., 2019; Ukkola et al., 2016). However, the effects remain largely debated, with recent observation-constrained modeling suggesting that the direct regulation of CO_2 on global streamflow via vegetation physiology may be buffered or relatively weak compared to precipitation (H. Wei et al., 2024). Studies also show that runoff is highly sensitive to vegetation change (Goulden & Bales, 2014; S. Zhou et al., 2023). Concurrently, vegetation changes have a profound impact on soil water and affect streamflow at multiple temporal and spatial scales (Zhang, Viglione, et al., 2022). Furthermore, regional evidence suggests that the climate and vegetation impacts are highly heterogeneous. For instance, a 1% reduction in precipitation in southeastern Brazil leads to a wide range of decreases in streamflow, varying from 1.83% to

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4.75% (Deusdará-Leal et al., 2022). The impact of vegetation change on streamflow is higher in arid regions than in humid ones (Yang et al., 2022; G. Zhou et al., 2015). Meanwhile, temperature and snow depth also affect streamflow variations in snowy regions (Xu et al., 2024).

To conclude, most of these large or regional-scale assessments focus on changes in streamflow magnitude, including extreme high and low flow (Guzha et al., 2018), and baseflow (Tan et al., 2020). Streamflow characteristics, including magnitude, frequency, duration, and temporal variations, should be considered along with other hydrological variables to provide a more comprehensive quantification of the climate and vegetation influences (H. K. McMillan, 2021; Poff et al., 1997). However, streamflow signatures of streamflow duration and timing are frequently ignored, despite their important roles in the ecological environment. In fact, regional studies highlight the importance of focusing more on the responses of streamflow timing and duration signatures since they show larger responses compared to streamflow magnitude signatures (Huang et al., 2022, 2024). Therefore, as the climate drought could propagate to hydrological drought (Van Loon, 2015), quantifying the propagation of environmental change to multi-dimensional streamflow signatures is of great importance. This requires a comprehensive study on the impact of meteorological and vegetation drivers on multiple aspects of hydrological processes at a large spatial scale. Particular attention is how the climate seasonality transmits to streamflow seasonality differently across varying aridity catchments (Gnann et al., 2020). A critical knowledge gap emerges regarding how the changes in meteorological and vegetation factors propagate through hydrological pathways, and alter different streamflow signatures. Specifically, it remains unclear whether these changes are damped or amplified during their propagation and increasing temporal scales, and how these mechanisms vary across different climate zones.

To fully investigate these mechanisms, it is necessary to use a robust hydrological modeling framework. While land surface models offer highly mechanistic representations of vegetation-hydrology interactions (Lawrence et al., 2019; Sellers et al., 1996), conceptual coupled models such as HBV-PML offer practical advantages in terms of computational efficiency, lower data requirements for parameterization, and broad applicability in large-scale applications, while still capturing the essential dynamics of vegetation evapotranspiration (Huang et al., 2022, 2024). However, due to the need for flood forecasting, traditional streamflow calibration often neglects the performance of low flows. As high-flow simulations improve, low-flow accuracy is frequently compromised (Euser et al., 2013). For example, commonly used streamflow simulation metrics such as the Nash-Sutcliffe Efficiency (Nash & Sutcliffe, 1970, p. 1) and Kling-Gupta Efficiency (KGE) (Gupta et al., 2009) emphasize long-term averages and high flows, often at the expense of low-flow characteristics, limiting their ability to comprehensively capture the full range of streamflow signatures. Studies show that hydrological models constrained by multi-objective functions can effectively identify and reduce errors and uncertainties, achieving balanced performances through the algorithm (Efstratiadis & Koutsoyiannis, 2010). Incorporating streamflow signatures as part of objective functions of model calibration shows improved performance compared with the traditional hydrological modeling that only uses streamflow magnitude constraints (Melišová et al., 2020). Therefore, incorporating a variety of streamflow signatures, which are not limited to magnitude-based signatures, can provide richer information, impose better constraints on hydrological models, and offer a robust quantitative description of uncoupling the drivers of streamflow changes (H. McMillan, 2020).

To address the above challenges, this study utilized a hydrological model integrating dynamic vegetation processes, leveraging multiple streamflow signatures as model constraints to derive a robust quantification of climate and vegetation influences. To isolate natural responses, catchments highly influenced by human activities (total reservoir maximum capacity >10% of mean annual streamflow) were excluded here (Huang et al., 2022). Changes in precipitation, temperature, and remote sensing Leaf Area Index (LAI) are selected to represent climate and vegetation disturbance. Since the observed LAI is used to capture actual structural vegetation changes, it also inherently includes the historical structural change due to CO_2 fertilization. Given its high complexity, decoupling the physiological and structural effects of CO_2 falls beyond the scope of the current study.

The specific objectives of this study are:

1. To improve the simulation of multi-dimensional streamflow signatures at large scales using a hydrological model that incorporates dynamic vegetation processes;
2. To quantify how changes in meteorological and vegetation factors propagate through hydrological pathways and ultimately alter streamflow signatures across different climate zones, especially to examine if the environmental changes are amplified or damped through the propagation effects; and

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3. To identify the dominant drivers of changes in streamflow signatures globally, and compare their spatial patterns across different climate zones.

2. Materials and Methods

2.1. Data and Study Area

Daily streamflow time series were compiled from multiple sources (Text S1, Figure S1 and S2 in Supporting Information S1), including but not limited to current well-maintained and open-sourced hydrological data sets (Addor et al., 2017a, 2017b; Alvarez-Garreton et al., 2018a, 2018b; Chagas et al., 2020a, 2020b; Coxon et al., 2020a, 2020b; Fowler et al., 2021, 2024). The quality control methods (Gudmundsson et al., 2016, 2018; Klein Tank et al., 2009) were thoroughly conducted for the entire data set. From the merged data set, we selected 2,252 catchments to robustly assess the influence of climate and vegetation changes on hydrological processes (Figure 1). Selection criteria required high-quality streamflow records covering at least 80% of the 1982–2018 period and minimal anthropogenic regulation, defined by a maximum reservoir capacity of less than 10% of the mean annual streamflow. The 2,252 catchments cover over 17% of the global land area and span the major Köppen climate zones, including 285 equatorial, 97 arid, 1,094 warm-temperate, 749 snow, and 27 polar catchments.

The meteorological (precipitation and temperature) and vegetation (LAI) variables of each catchment were derived from MSWEP V2.8 (Beck et al., 2019), MSWX (Beck et al., 2022), GLASS V6 (Ma & Liang, 2022), NOAA globally averaged marine surface monthly mean CO_2 concentration data (Global Monitoring Laboratory, 2026; Lan et al., 2026), respectively, and were spatially averaged over each catchment (Figure S5 in Supporting Information S1).

2.2. Methods

2.2.1. HBV-PML

The HBV-light (Hydrologiska Byråns Vattenbalansavdelning) model is a conceptual hydrological model with simplicity, limited parameter requirements, fast computation, and reliable performance (Bergström, 1995, 1995, 1995; Bergstrom & Lindstrom, 2015; Seibert, 1997). However, it relies on potential evapotranspiration as one of the inputs, which fails to fully account for the impact of vegetation changes on the evapotranspiration process. To incorporate the mechanistic simulation of vegetation changes on hydrological processes, a carbon-water coupling evapotranspiration model, PML-V2 (Y. Zhang et al., 2019), is used to replace the evapotranspiration module of the HBV conceptual hydrological model, resulting HBV-PML model (Huang & Zhang, 2026). PML-V2 partitions evapotranspiration into transpiration (Et), canopy interception (Ei), and soil evaporation (Es) based on physically meaningful drivers (precipitation, temperature, CO_2 concentration, LAI, radiation, etc.), enabling the model to mechanistically capture the propagation of climate and vegetation changes to streamflow. The key functions of PML-V2 are derived from classical theories (Farquhar et al., 1980; Gan et al., 2018; Morillas et al., 2013; Thornley, 1976; van Dijk & Bruijnzeel, 2001; Yu et al., 2004; Zhang et al., 2010, 2016) and can be found in Text S2. The coupling processes are as follows:

1. The three evapotranspiration components from PML-V2 are restructured into two groups: Ei_{PML} (interception evaporation) and Est_{PML} (the sum of soil evaporation and plant transpiration, $Est_{PML} = Es_{PML} + Et_{PML}$).
2. For Ei, its value is constrained by precipitation to simulate canopy interception processes. When $P < Ei_{PML}$, $Ei = P$, and only snowmelt water enters the soil. When $P > Ei_{PML}$, $Ei = Ei_{PML}$, net precipitation ($P' = P - Ei$) and snowmelt enter the soil layer.
3. For Est, this study constrained its magnitude using both the water entering the soil layer (P' + melt) and the soil moisture storage (SM), reflecting the dynamic process of soil moisture entering the atmosphere through evaporation and plant transpiration. If P' + melt + SM < Est_{PML}, then Est = P' + melt + SM. Otherwise, $Est = Est_{PML}$, and the remaining water (P' + melt + SM−Est_{PML}) enters the upper groundwater zone (UZ).
4. Finally, the actual evapotranspiration (AET) is determined by the sum of the constrained Ei and Est, that is, AET = Ei + Est.

Through the model coupling, the HBV-PML model retains the original HBV soil moisture routine while integrating the process-based, vegetation-responsive evapotranspiration simulation from PML-V2. The model inputs include precipitation, LAI, temperature (mean, minimum and maximum temperature), short-wave radiation, long-wave radiation, albedo, emissivity, specific humidity, and CO_2 concentration. Notably,

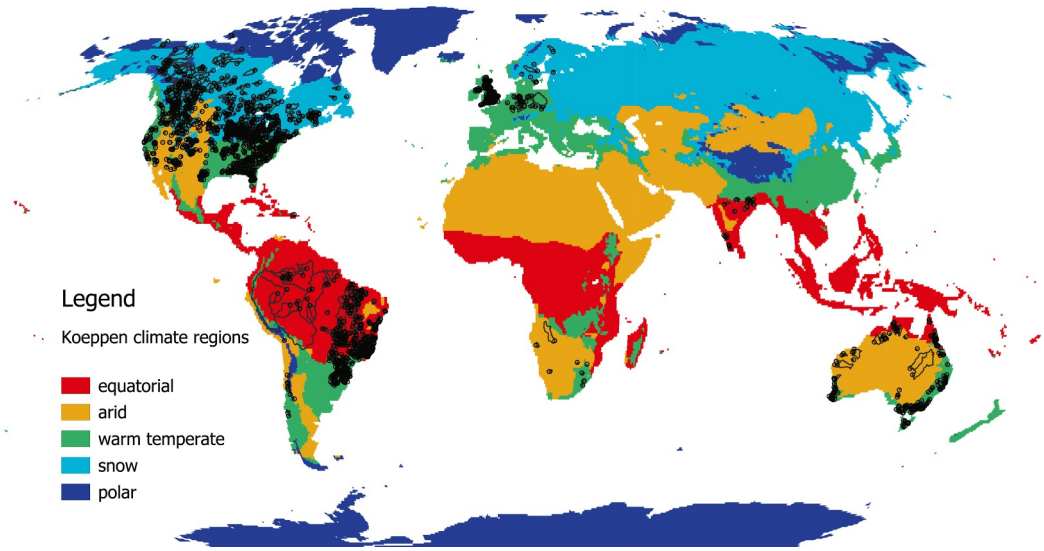


Figure 1. Distribution of studied catchments and their hydrological stations.

intermediate variables such as net radiation, vapor pressure deficit and equilibrium soil evaporation are dynamically generated from these primary inputs, enabling physically consistent responses during simulations. As illustrated in Figure 2, this model structure allows for a more mechanistic representation of vegetation effects on watershed hydrology, particularly in terms of dynamic responses such as vegetation interception, soil

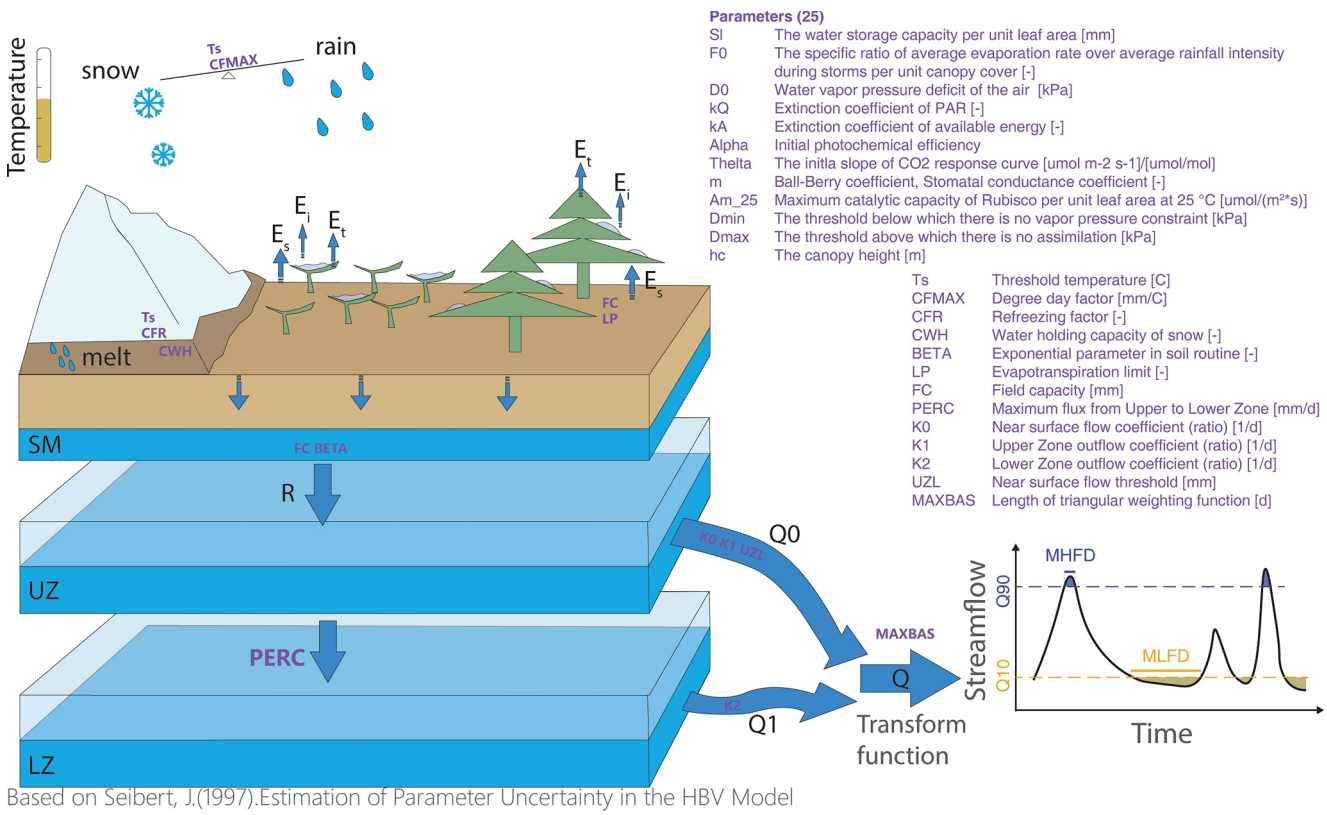


Figure 2. Structure of HBV-PML. The temporal variables are shown in black text, and the parameters are shown in purple text. The parameters of evapotranspiration are listed in the top right panel but not labeled in the conceptualized plot for better viewing. The units of water fluxes are mm day^{-1}, while the unit of temperature is in Celsius (°C). Four typical streamflow signatures are identified in the diagram.

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Table 1
Parameter Ranges of the HBV-PML Model

Parameters	Module	Lower limits	Upper limits
Alpha	Evapotranspiration	0	0.048
Thelta		0	0.069
m		0	25.755
Am_25		0	46.375
D0		0.501	2
kQ		0.1	1
kA		0.68	0.899
S_sls		0	0.174
fER0		0	0.158
VPDmin		0.657	1.492
VPDmax		3.5	6.5
hc		0.01	10
Ts	Snow melt	−5	5
CFMAX		0	20
CFR		0	1
CWH		0	0.8
BETA	Soil moisture	0	7
LP		0.3	1
FC		1	2000
PERC	Responding function	0	100
K0		0.05	2
K1		0.01	1
K2		0.02	0.1
UZL		0	300
MAXBAS	Routing	1	10

evaporation, and plant transpiration. Thereby it improves the model's ability to simulate responses to vegetation dynamics. In this framework, soil moisture is directly depleted by AET, which in turn regulates groundwater recharge and runoff generation through the response function. Finally, runoff is routed via a triangular weighting function (MAXBAS) to generate streamflow at the outlet, which can further calculate simulated streamflow signatures. Parameter equifinality remains a persistent challenge in hydrological modeling, since it can yield realistic streamflow simulations while masking uncertainties in internal processes (Moges et al., 2021). Therefore, the simulated AET are evaluated against the PML-V2 product across five climate regions (Figure S9 in Supporting Information S1) using a multi-dimensional performance evaluation index (Taylor, 2001). The strong agreement indicates that the uncertainty has been effectively reduced by physically constraining the evapotranspiration process.

There are 25 parameters in HBV-PML. Table 1 shows the parameters and their ranges, according to the definitions of HBV (Seibert & Bergström, 2022) and PML-V2 model (Y. Zhang et al., 2019).

We selected eight streamflow signatures representing different aspects of streamflow signatures including magnitude, frequency, duration, and dynamics (Table 2). These signatures comprehensively capture high flow, average state, and low flow conditions, ensuring strong representativeness. Furthermore, to avoid outliers in extreme conditions such as the 99th percentile streamflow (H. McMillan et al., 2023), this study uses the 90th and 10th percentile flow signatures, to reduce the influences of outliers and improve the robustness of the analysis.

2.2.2. Objective Function

The traditional objective function is set as the baseline to optimize KGE_Q (Gupta et al., 2009) in this study (F_2). Subsequently, we developed a multi-streamflow signature objective function (F_1) to evaluate the overall performance. In F_1, weights of the performance of streamflow signatures are assigned as follows: 50% to daily mean streamflow, 30% to log-transformed Quantile_10, and 10% to Quantile_90 and LFD_Q10, respectively. This weighting strategy ensures a balanced emphasis on streamflow magnitude and variability, thus improving the robustness of model calibration beyond a single performance metric. It should be noted that Abias is adopted to avoid NA values of KGEs due to the occurrence of -Inf in absolute deviation. Furthermore, to emphasize low-flow conditions, Quantile_10 values

Table 2
Streamflow Signatures in This Study

Type	No	Streamflow signatures	Streamflow signatures (full name)	Unit	Description
Magnitude	1	Quantile_90	Quantile 90th of the streamflow	mm day^{-1}	High flow
	2	Quantile_10	Quantile 10th of the streamflow	mm day^{-1}	Low flow
	3	Mean	Mean of the streamflow	mm day^{-1}	Mean flow
Dynamic	4	FDCs	Slope of Flow Duration Curve	–	The slope between the 33rd and 66th flow duration curve
Frequency and duration	5	LFD_Q10	Low Flow Days of streamflow less than the threshold of mean annual Quantile_10	Days (of a year)	Frequency of low flows
	6	HFD_Q90	High Flow Days of streamflow more than the threshold of mean annual Quantile_90	Days (of a year)	Frequency of high flows
	7	MLFD_Q10	Maximum Durations of consecutive LFD_Q10	Days (of a year)	Extreme low flows
	8	MHFD_Q90	Maximum Durations of consecutive HFD_Q90	Days (of a year)	Extreme high flows

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were log-transformed prior to calculation. The threshold for LFD_Q10 was determined by the Quantile_10 values of the observed time series, ensuring a consistent benchmark for comparison.

The objective function of scheme one is:

$$F_1 = \min\left\{\left(1 - \mathrm{KGE}_Q\right) * 0.5 + \left(1 - \mathrm{KGE}_{Q90}\right) * 0.1 + \mathrm{Abias}_{\log(Q10)} * 0.3 + \mathrm{Abias}_{\mathrm{LFD}(Q10)} * 0.1\right\} \quad (1)$$

The objective function of scheme two is:

$$F_2 = \min\left(1 - \mathrm{KGE}_Q\right) \quad (2)$$

$$\mathrm{KGE} = 1 - \sqrt{(r-1)^2 + \left(\frac{\sigma_m}{\sigma_o} - 1\right)^2 + \left(\frac{\mu_m}{\mu_o} - 1\right)^2} \quad (3)$$

$$\mathrm{Abias} = \frac{\sum_{i=1}^{n} \left| y_i - \hat{y}_i \right|}{\sum_{i=1}^{n} \left| y_i \right|} \quad (4)$$

where r represents the coefficient of correlation, σ_m and σ_o are the simulated and observed standard deviation, μ_m and μ_o represents mean values of observed and simulated data, respectively, n refers to the total temporal periods. i represents the ith temporal periods, $\hat{y}_i$ and y_i are simulated and observed data, respectively. The range of KGE is $(-\infty, 1]$. The best values of KGE and Abias are 1 and 0, respectively.

2.2.3. Impacts Quantification

To disentangle the effects of meteorological and vegetation drivers on hydrological processes and streamflow signatures, we designed five sensitivity experiments using calibrated parameters from the 2,252 catchments. Both dynamic and static scenarios were constructed for vegetation and climate forcing (Table 3). To minimize long-term trends and noise in the "static" inputs while retaining data variability (Hyndman & Killick, 2024; Wen et al., 2019), the annual trends for each variable were removed. In static scenarios, precipitation and LAI were scaled proportionally to preserve their annual distribution. In contrast, temperature was adjusted via mean-shifting to avoid the physical inconsistencies inherent in proportional scaling. To ensure physical consistency of detrended scenarios of S1 ~ 4, the internal variables (i.e., vapor pressure deficit, net radiation, equilibrium soil evaporation. etc.) are also updated along with the detrended inputs. Streamflow simulations were conducted for the period 1982–2018 under each scenario, following a three-year warm-up period of 1979–1981. Comparing simulations across these scenarios enabled the investigation of quantified mechanisms and attribution of hydrological changes to vegetation and meteorological variables.

Annual streamflow signatures were derived from the simulated streamflow series for each scenario. The relative contribution of each driving factor was quantified by calculating the percentage difference between the scenario outputs (S_i) and the baseline (S_0) simulation, expressed as follows:

$$CR = \frac{s_0 - s_i}{s_0} \times 100 \quad (5)$$

This approach facilitates the attribution of the meteorological and vegetation impacts on key hydrological components (e.g., snowmelt, soil moisture, and streamflow signatures, etc.). Consequently, the impacts of changes in precipitation, temperature, and LAI were decomposed to reveal their respective contributions in altering hydrological processes.

Figure 3 illustrates the multi-year average percentage differences between the original and detrended inputs over the study period. Precipitation patterns (Figure 3a) exhibit strong spatial heterogeneity with coherent regional patterns. Notably, western North America and southern Australia exhibit lower observed precipitation compared to the detrended values, indicating long-term drought trends likely driven by climate change. Temperature differences (Figure 3b) are predominantly positive worldwide, reflecting pervasive warming. Exceptions include negative anomalies in northern North America, which correspond to sub-zero baseline temperatures. The

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Table 3
Scenario Design of Quantifying the Impact of Climate and Vegetation Change on Streamflow Signatures

Scenario	Experimental setup	P	T	LAI
0	Benchmark	Observed data	Observed data	Observed data
1	Detrended LAI	Observed data	Observed data	Scale-detrended
2	Detrended P	Scale-detrended	Observed data	Observed data
3	Detrended T	Observed data	Additive shifting-detrended	Observed data
4	Detrended P, T and LAI	Scale-detrended	Additive shifting-detrended	Scale-detrended

Note. Scale-detrended refers to detrended at an interannual scale and proportionally scaled within the year. Additive shifting-detrended refers to detrended at an interannual scale and adjusted daily values by additive shifting within the year.

localized cooling trends in northern Australia are consistent with observations of the Australian Bureau of Meteorology (Englart, 2019). LAI differences (Figure 3c) highlight spatially heterogeneous changes in LAI based on the GLASS data set. Vegetation browning is evident in western North America and southeastern Australia, whereas most other regions exhibit greening trends. Overall, Figure 3 highlights distinct spatial patterns in precipitation, temperature, and LAI changes across 2,252 unregulated catchments when comparing original and detrended conditions.

2.2.4. Stratified Statistics

While the use of 2,252 unregulated catchments allows for a robust, large-sample quantification of regional propagation mechanisms, large data sets can inherently obscure subtle input-output relationships. To address this, we employed a stratified analysis approach. By partitioning changes in climate or vegetation into quantile-based

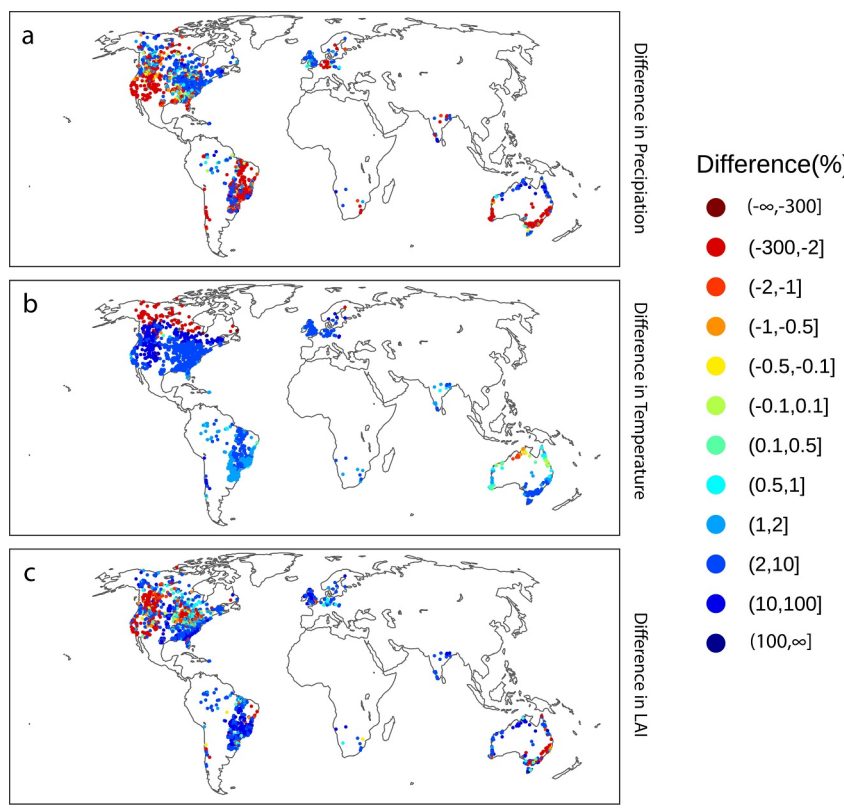


Figure 3. Percentage differences between observed and trend-removed input variables. The observed precipitation, temperature and leaf area index are obtained from MSWEP, MSWX, and GLASS data sets, respectively.

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subgroups, this method facilitates the characterization of systematic patterns while mitigating the influence of outliers. The method has been well-suited for large-sample hydrological analyses (Kleinbaum et al., 2013).

Specifically, changes in key driving variables were grouped into five groups using 20-percentile intervals. Variations in hydrological responses across these groups were then compared to explore how each variable change influences hydrological processes. To further enhance interpretability and generalizability, the stratified method was also conducted independently for each climate zone.

2.2.5. Absolute Contribution Analysis

To quantify the individual influence of multiple driving variables, this study applied an absolute contribution analysis. Based on multi-scenario simulations, this approach decomposes the total change into the independent effects of precipitation, temperature, LAI, and their interactions (Equation 6). This allows for a robust assessment of the relative impact of each variable on key hydrological components and streamflow signatures.

$$C_i = \frac{|\Delta \mathrm{Var}_i|}{\sum_{i=1}^{n} |\Delta \mathrm{Var}_i|} \tag{6}$$

where C_i is the absolute contribution of the ith variable, $\Delta \mathrm{Var}_i$ is the output change due to input change of the ith variable and the sum of individual contributions is 1.0. For example, the absolute contribution of precipitation to certain variable can be calculated as follows:

$$\begin{aligned} C_p &= \frac{|\Delta P|}{|\Delta P| + |\Delta \mathrm{LAI}| + |\Delta T| + |\Delta \mathrm{Interaction}|} * 100 \\ &= \frac{|S0 - S2|}{|S0 - S2| + |S0 - S1| + |S0 - S3| + |S4 - S1 - S2 - S3|} * 100 \end{aligned} \tag{7}$$

During the preparation of this manuscript, Gemini 3.1 Pro was used to improve the spelling, grammar, clarity, and readability of the manuscript. After using this tool, the authors carefully reviewed and edited the text and took full responsibility for the final content.

3. Results

3.1. Model Performance in Capturing Diverse Streamflow Signatures Across Five Climatic Zones

The HBV-PML model, incorporating vegetation dynamics, yields robust simulations of hydrological processes under different calibration schemes. In Scheme 1, median KGE values reached 0.65 and 0.53 for the calibration and validation periods, respectively. The corresponding values are 0.57 and 0.44 for Scheme 2 (Figure S6 in Supporting Information S1). As for individual streamflow signatures, the HBV-PML model performs better for mean flow (Mean) and high flow (90th percentile), compared to low flow (10th percentile) and streamflow duration signatures (Figure S7 in Supporting Information S1). Notably, Scheme 1 especially improves the performance of low-flow days, underscoring the benefits of the weighted calibration scheme.

Given that hydrological models generally exhibit higher performance in humid regions compared to arid regions, benchmarking efficacy across climate zones is a prerequisite for mechanistic analysis. Figure 4 illustrates the KGE differences between Scheme 1 and Scheme 2 across the five Köppen climate zones for both calibration and validation periods. Positive values denote the superiority of Scheme 1 over Scheme 2. In arid regions, Scheme 1 exhibits a clear advantage, with median KGE differences exceeding ~0.2 during both calibration and validation periods, suggesting improved performance in capturing the overall performance of streamflow signatures. In polar regions, Scheme 1 also shows slightly better performance. In equatorial, warm temperate and snow regions, the advantages are smaller, which might indicate a good performance in Scheme 1 as well. Consequently, due to its consistently superior global performance, the parameters of Scheme 1 were selected for the subsequent analysis. Additionally, we observed that model uncertainty amplifies with increasing aridity, as evidenced by wider dispersion ranges in the simulations.

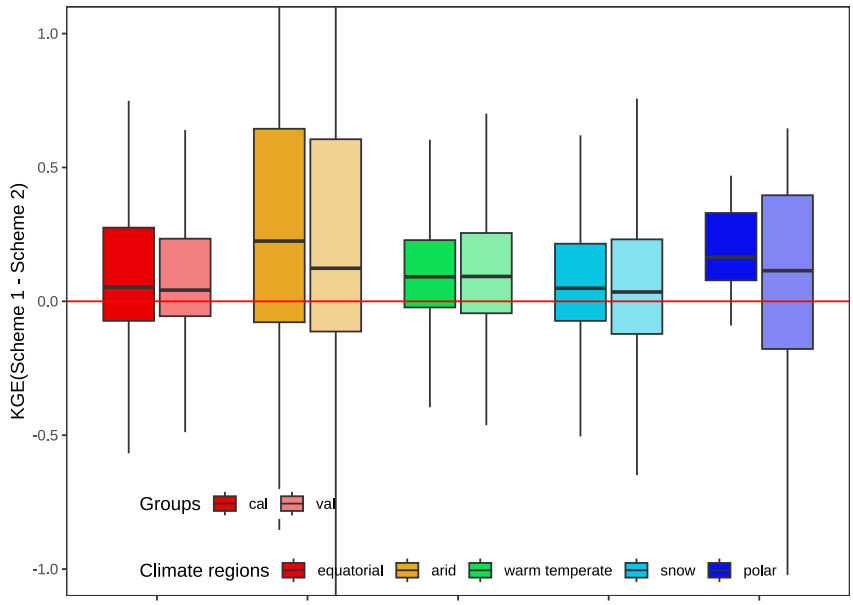


Figure 4. Comparison of different calibration schemes using Kling-Gupta efficiency values across all streamflow signatures.

3.2. Divergent Sensitivities of Streamflow Signatures to Environmental Changes in Five Climate Zones

The sensitivity of streamflow signatures to climate and vegetation changes across different climatic regions is analyzed using a stratified statistical approach. In Figure 5, each point represents the median response within each quintile bin (20th-percentile interval) of the corresponding input variable, while the fitted lines depict the sensitivity of streamflow signatures to driver changes across climate zones. Figure 5 illustrates how magnitude and dynamic signatures, as well as frequency and duration signatures, respond to variations in precipitation, temperature, and LAI.

Changes in LAI and precipitation exhibit a wider range in equatorial and warm temperate regions, whereas changes in temperature show greater variability in arid and polar regions. The influences of vegetation change (Change_LAI) on streamflow magnitude and dynamic signatures are relatively consistent across climatic zones. However, with LAI increasing, the decline in Quantile_10 and Mean is more pronounced in polar and snow-dominated regions compared to other climate regions, suggesting a stronger vegetation control on hydrological processes in cold environments. Conversely, high-flow signatures (MHFD_Q90 and HFD_90) show greater sensitivity to LAI changes in arid regions, suggesting greening is more effective at attenuating flood peaks in drylands. Studies also reveal a stronger response of LAI increase to runoff in arid regions (e.g., mean streamflow in this study), showing significant runoff reductions due to LAI increases in global drylands, specifically 48.0%, 33.2%, and 20.2% in arid, semi-arid, and sub-humid regions (Luo et al., 2020).

The effect of precipitation change (Change_P) on streamflow magnitude and dynamics also displays a broadly consistent pattern across climatic zones. Nevertheless, the response is markedly more sensitive in polar regions, where steeper slopes indicate that increases in precipitation lead to a more pronounced enhancement of streamflow signatures. This agrees with the recent findings of increased winter precipitation enhancing peak discharge in spring and early summer in arctic rivers (Park et al., 2024). Additionally, the slope of mean streamflow (second row, third column of Figure 5) to precipitation is notably lower in arid zones compared to other regions. This supports previous studies demonstrating that runoff is less sensitive to precipitation in drylands (Berghuijs et al., 2017).

In comparison, hydrological responses to temperature change (Change_T) are more complex and climate-region dependent. In snow and polar climates, the decline in high-flow (Quantile_90) and a concurrent increase in low-flow (Quantile_10) are related to rising temperatures. However, such effects are limited or negligible in other regions. Mean streamflow generally decreases with warming across most regions. Notably, in polar regions, streamflow exhibits a nonlinear response to temperature change, where it tends to increase streamflow in the

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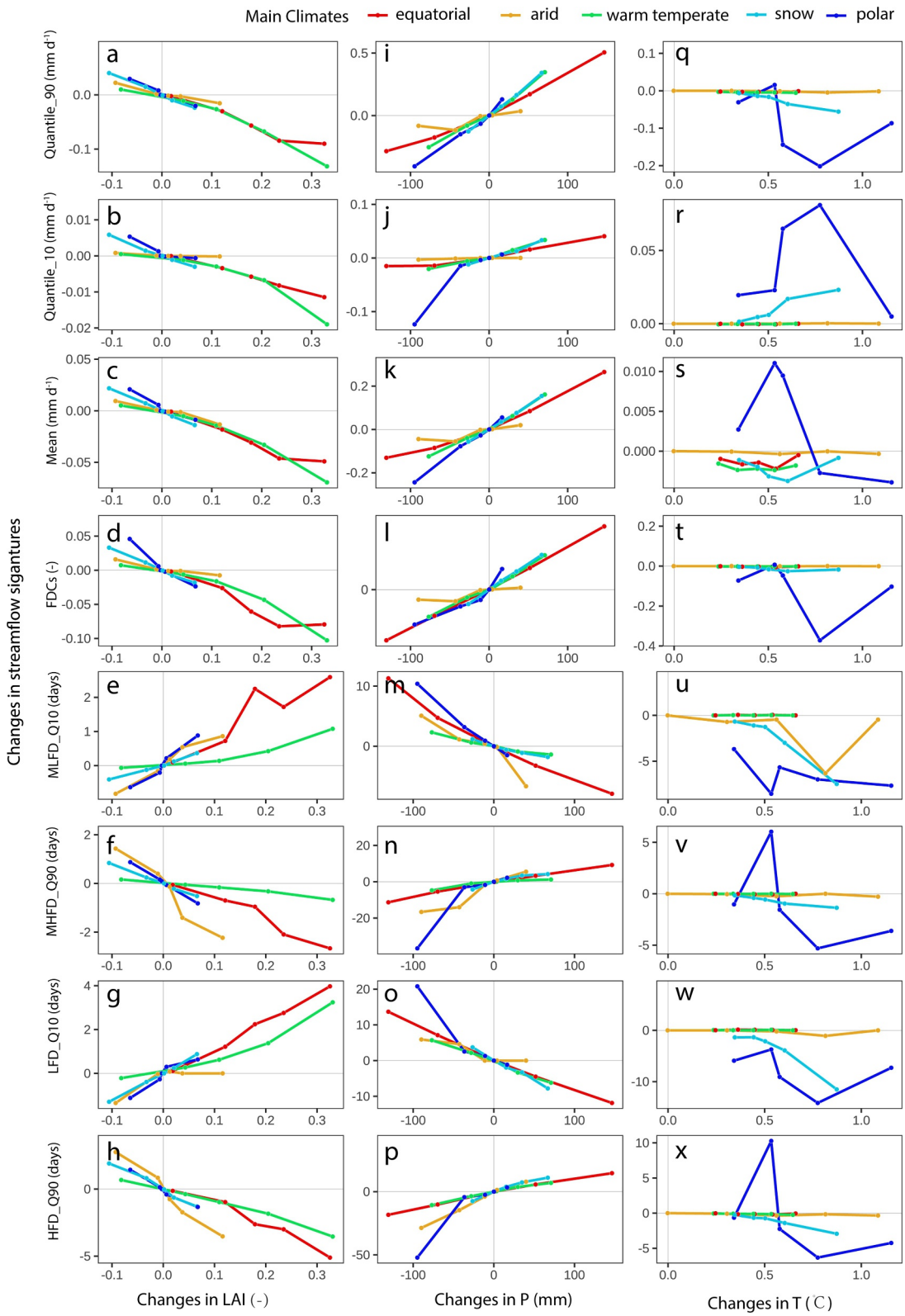


Figure 5. Response of streamflow signatures to changes in leaf area index (LAI), P, and T according to different climate regions. Panels (a–h) are for responses to LAI changes; panels (i–p) are for responses to P changes; panels (q–x) are for responses to T changes. Curves denote sensitivities derived from a robust stratified approach across five climate regions. Catchments were categorized into quintiles based on historical input changes (ΔP, ΔT, or ΔLAI). Points represent the median input and output changes for each quintile group.

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moderate warming condition, while higher levels of warming result in a decline. This pattern likely reflects the combined effects of reduced persistent snowpacks (Mohammadzadeh Khani et al., 2022) and uncertainties associated with the relatively small sample size from polar basins.

In summary, the relationships between changes in climate and vegetation and variations in streamflow signatures exhibit broadly consistent patterns across most climate zones, with noticeable exceptions in certain climate zones. Streamflow signatures in polar regions are particularly sensitive to temperature changes, reflecting the strong thermal control on snow- and ice-dominated hydrological regimes. In terms of overall magnitude in *y*-axis, precipitation-induced change is the largest for each streamflow signature, underscoring its dominant role in changes of streamflow signatures. Streamflow signatures are more sensitive to LAI change in the arid zone (Luo et al., 2020), compared to each-unit changes in precipitation or temperature (Table S4 in Supporting Information S1). Temperature change generally shows a weaker influence compared with the other two variables.

3.3. Propagation Effects of Climate and Vegetation Change on Hydrological Processes Derived From Five Main Climate Zones

By summarizing the sensitivity (i.e., rate of response variable change to input variable change as slopes of Figure 5) of key hydrological variables and streamflow signatures on each one unit increase of LAI, precipitation and temperature in different climate regions (Table S4 in Supporting Information S1), a quantified conceptual map of the change propagation mechanism is constructed in Figure 6. Illustrations of typical catchment behaviors are shown in Figures S12–S14 in Supporting Information S1.

As shown in Figure 6a, a one-unit increase in LAI enhances both canopy interception (8.81 ± 1.92 mm yr^{-1}) and vegetation transpiration (38.97 ± 25.31 mm yr^{-1}), while concurrently reducing soil evaporation (-25.51 ± 22.36 mm yr^{-1}). This shift results in a slight increase in total evapotranspiration, leading to a reduction in soil moisture (-30.59 ± 14.75 mm yr^{-1}). Consequently, both upper and lower soil water storage levels decline (-0.12 ± 0.11 mm yr^{-1} and -1.37 ± 0.74 mm yr^{-1}, respectively). The overall decrease in soil moisture availability contributes to reductions in streamflow magnitude signatures, including Quantile_90, Quantile_10, mean flow, and flow duration curves (FDCs). Correspondingly, the frequency and duration of high-flow events decrease (-9.19 ± 6.18 days yr^{-1} and -4.64 ± 4.66 days yr^{-1}, respectively), whereas those of low-flow events increase (5.44 ± 1.18 days yr^{-1} and 4.69 ± 3.06 days yr^{-1}, respectively), indicating a general decline in water availability with vegetation greening. As LAI increases, AET increases and soil moisture decreases (Chen et al., 2024; Liu et al., 2025), particularly in dry regions (Zeng et al., 2018). Studies also show that greening tends to boost canopy interception and transpiration, shifting more precipitation into vegetation-mediated evapotranspiration and reducing soil moisture, especially in Central Asia, Central Africa, and southern Australia (Liu et al., 2025). Although this study employs LAI as the key variable to represent vegetation, LAI may not fully represent vegetation changes in terms of changes in land-cover and vegetation structure (e.g., rooting depth, phenology). Vegetation cover changes contribute to a $30.7\% \pm 22.5\%$ change in runoff variations in forest regions (X. Wei et al., 2017), while land cover change is limited in certain areas (Huang et al., 2022), its influence on global hydrology requires further investigation. Similarly, evaluating the hydrological impact of vegetation structure changes (Piao et al., 2020) relies on more detailed data and model structures from ecosystem models. Future research should incorporate these additional details to better quantify their specific effects on hydrological processes.

In Figure 6b, one unit increase in precipitation (0.54 ± 0.12 mm yr^{-1}) enhances snowmelt (0.02 ± 0.26 mm yr^{-1}) and elevates total evapotranspiration along with its components (0.14 ± 0.14 mm yr^{-1}, 0.01 ± 0.01 mm yr^{-1} and 0.2 ± 0.16 mm yr^{-1} for Ei, Et and Es, respectively). However, as the increases in rainfall and snowmelt exceed the increase in evapotranspiration, soil moisture content increases (0.07 ± 0.11 mm yr^{-1}). Consequently, improved soil moisture conditions lead to increases in streamflow magnitude signatures, including Quantile_90 (0.002 ± 0.002 mm yr^{-1}), Quantile_10 (0.0001 ± 0.0006 mm yr^{-1}), mean flow (0.001 ± 0.001 mm yr^{-1}) and slope of flow duration curves (FDCs, 0.002 ± 0.001 years^{-1}). Correspondingly, the frequency and duration of high-flow events increase (0.09 ± 0.024 days yr^{-1} and 0.04 ± 0.17 days yr^{-1} for HFD and MHFD, respectively), while those of low-flow events decrease (-0.06 ± 0.09 days yr^{-1} and -0.04 ± 0.05 days yr^{-1} for LFD and MLFD, respectively), reflecting an overall enhancement in water availability. Apart from canopy interception evaporation and vegetation transpiration, the effects of precipitation change on streamflow signatures are opposite to those of the LAI change (Alkama et al., 2010; Deng et al., 2023). Comparing Figures 6a and 6b, it fits well with

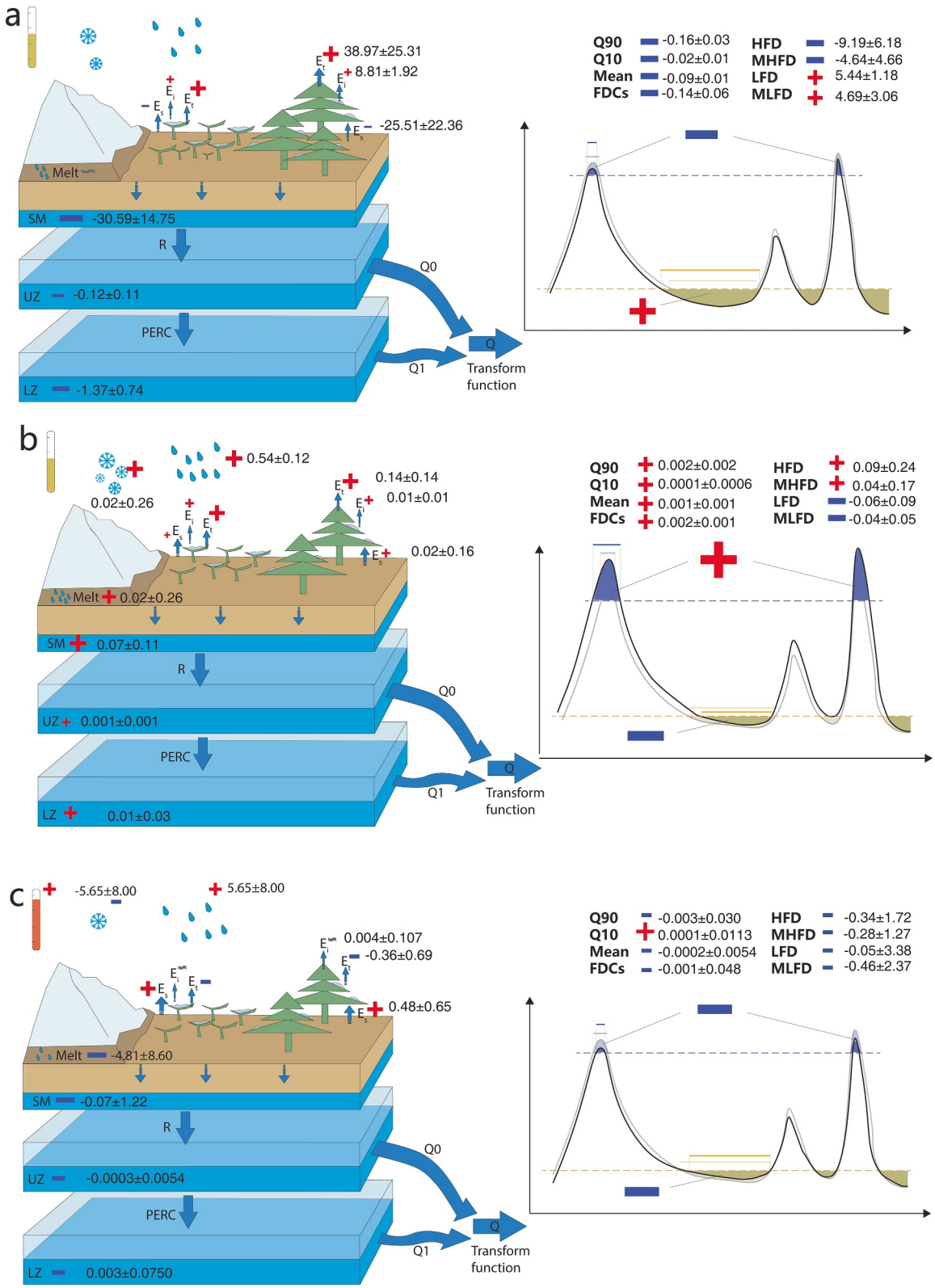


Figure 6. Conceptualized mechanisms of the influences of climate and vegetation change on hydrological processes. Panels (a–c) represent propagating effects of one-unit increase of leaf area index, precipitation and temperature, respectively. The texts represent the global median ± one standard deviation of the changes across the median values of five climate regions.

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the finding showing the rising P-induced Q increase is largely offset by greening-caused Q decline globally (Luo et al., 2020).

A primary mechanism by which warming alters hydrological processes is through changing the phase of precipitation, resulting in a higher proportion falling as rain rather than snow. This phase shift leads to a reduction in snowfall (-5.65 ± 8.00 mm yr^{-1}) and subsequent snowmelt (-4.81 ± 8.60 mm yr^{-1}) (Figure 6c). Simultaneously, although interception evaporation tends to decrease, the elevated soil evaporation (0.48 ± 0.65 mm yr^{-1}) along with the decreased vegetation transpiration (-0.36 ± 0.69 mm yr^{-1}) result in a net increase in total evapotranspiration. This enhanced moisture loss causes a decline in soil moisture (-0.07 ± 1.22 mm yr^{-1}), subsequently reducing both upper and lower soil water storage (-0.0003 ± 0.0054 mm yr^{-1} and 0.0003 ± 0.0750 mm yr^{-1}). The depletion of soil moisture contributes to a decline in streamflow magnitude signatures, including Quantile_90 (-0.003 ± 0.030 mm yr^{-1}), mean flow (-0.0002 ± 0.0054 mm yr^{-1}), and slope of flow duration curves (FDCs, (-0.001 ± 0.048 mm yr^{-1})). In contrast, Quantile_10 (0.0001 ± 0.0113 mm yr^{-1}) exhibits an increasing trend. In addition, the frequency and duration of high-flow and low-flow events decrease simultaneously, where HFD, MHFD, LFD and MLFD decrease by -0.34 ± 1.72 days yr^{-1}, -0.28 ± 1.27 days yr^{-1}, -0.05 ± 3.38 days yr^{-1}, -0.46 ± 2.37 days yr^{-1}, respectively. This result might be attributed to the warming-induced shift in precipitation phase during the snow accumulation period. Studies show that in cold regions, snow-dominated basins warming causes less snow accumulation and more melt (Arora et al., 2008; Kumar et al., 2022), producing higher winter flows, earlier spring peaks and reduced late-summer runoff, which in turn modifies soil moisture seasonality and the timing of evapotranspiration (Barnett et al., 2005). That is to say, more precipitation falls as rain due to temperature increasing, thereby enhancing streamflow during the low flow periods, especially in winter. Overall, temperature-induced hydrological changes lead to a simultaneous reduction in high flows and an increase in low flows, indicating a weakening of hydrological extremes by equalization in the entire streamflow regime. Studies have also shown that the warming globe increases AET (Qiao et al., 2023) and snowmelt, which reduces soil moisture and alters streamflow regimes (Barnett et al., 2005).

3.4. Amplified Effects in Increasing Temporal Scales and Hydrological Processes

To examine the hydrological responses to climate and vegetation change with increased temporal scales, the model simulation period is divided into four sub-periods: Period 1 (1982–1991), Period 2 (1992–2001), Period 3 (2002–2011), and Period 4 (2012–2018). In order to compare across hydrological variables, the percentage form of hydrological responses is presented, and the key hydrological components are arranged according to the hydrological flux pathway in each row of Figure 7. Within subplot of each component, the temporal periods are also arranged along increasing temporal scales.

We observed a clear amplification of effects over time. For instance, increases in LAI drive progressively intensified increases in Ei and decreases in Es. Precipitation variability also increases over time, though the global median percentage change remains close to zero. However, a secondary amplification effect occurs during the transfer from actual evapotranspiration to streamflow signatures. For instance, increasing temperature leads to more pronounced reductions in snowfall and snowmelt, which in turn causes greater increases in Quantile_10 and more substantial decreases in LFD_Q10 and MLFD_Q10.

The combined effects of LAI, precipitation, and temperature changes are overall similar to the influence of precipitation alone. However, the widespread changes across multiple variables underscore the necessity of isolating the effects of individual drivers for a more thorough understanding of hydrological processes.

In summary, the propagation effects of LAI, P and T demonstrate amplified impacts in both increased temporal scales and hydrological processes, including total actual evapotranspiration, soil moisture dynamics, and streamflow signatures. The result underscores the complexity of hydrological system responses and highlights the potentially profound consequences of climate and vegetation change, particularly as effects of temporal accumulation and moisture transition mechanisms.

3.5. Attribution of Streamflow Signature Changes Across Five Climate Zones

This section further quantifies the relative contributions of precipitation, temperature, LAI, and their interactions to eight streamflow signatures across 2,252 global catchments using contribution analysis. The spatial contribution analysis for each catchment is shown as pie charts in Figures S15–S22 in Supporting Information S1.

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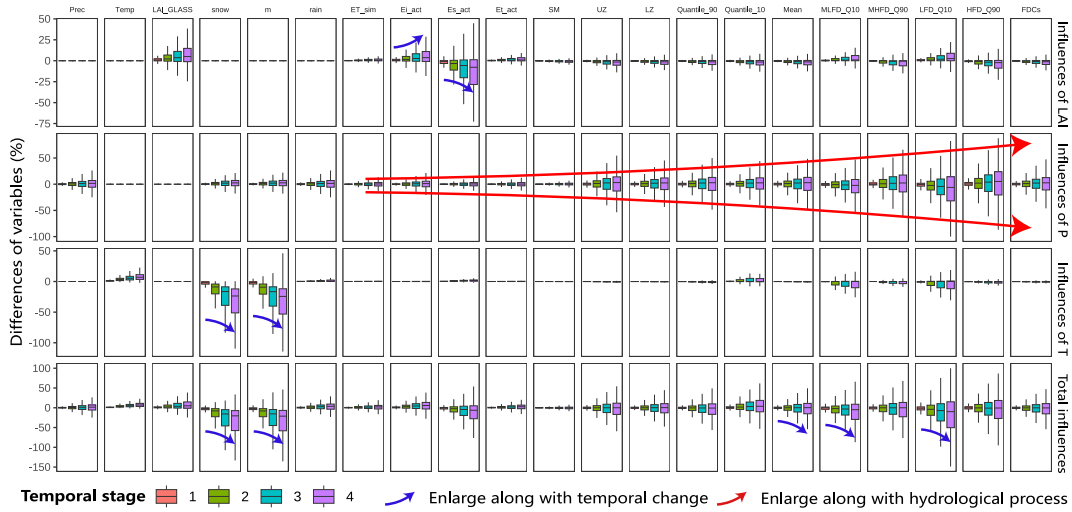


Figure 7. Propagation effects from changes in climate and vegetation variables to changes in key variables of hydrological process over four temporal stages.

Regarding dominant drivers (Table 4), precipitation is identified as the primary controlling factor for the eight streamflow signatures in 73.6 ± 10.5% of catchments, followed by LAI (13.7 ± 1.3%) and temperature (9.6 ± 8.5%). Notably, precipitation dominance is particularly pronounced for streamflow magnitude and high-flow signatures, including Mean, Quantile_90, and HFD_Q90. In contrast, temperature dominance is more prevalent in low-flow signatures such as Quantile_10, LFD_Q10, and MLFD_Q10, compared to mean and high flow signatures. This distinction suggests that climatic variables exert differentiated controls on specific streamflow signatures. Future studies are needed to further quantify the complex interactions between climate and vegetation change.

Apart from catchment numbers of dominant variables, the absolute contributions of climate and vegetation change to streamflow signatures in each catchment are also shown in Figures S15–S22 in Supporting Information S1, and are further summarized in Table 5.

Precipitation contributes more than 70% to Mean, Quantile_90, and HFD_Q90, which is higher than in other signatures. LAI also makes relatively high contributions to these three indicators (all exceeding 14%). Conversely, temperature exerts a markedly higher contribution to MLFD_Q10 compared to the other streamflow signatures. This aligns with findings in glacier-dominated catchments, where a 2.5% increase in mean streamflow has been attributable to rising temperatures (X. Wang et al., 2015). The interaction effects among precipitation, temperature, and LAI contribute more to MLFD_Q10 and MHFD_Q90 than to the other streamflow signatures, suggesting that streamflow duration-related signatures may involve more complex hydrological processes under a changing climate and vegetation.

Table 4
The Proportion of Catchments Dominated by Different Factors (%)

	Dominant by precipitation	Dominant by temperature	Dominant by LAI	Dominant by interaction
Quantile_90	79.6	6.5	13.9	–
Quantile_10	65.2	19.8	15.0	–
Mean	82.3	1.8	15.9	–
FDCs	77.0	9.5	13.5	–
MLFD_Q10	51.5	25.6	11.7	11.2
MHFD_Q90	71.3	9.9	12.1	6.8
LFD_Q10	68.2	19.6	12.1	0.1
HFD_Q90	80.0	5.8	14.0	0.2

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Table 5
The Summary of Absolute Contribution of Climate Change and Vegetation Change to Streamflow Signatures (%)

Streamflow signatures	Absolute contribution of precipitation	Absolute contribution of temperature	Absolute contribution of LAI	Absolute contribution of interaction
Quantile_90	71.0 ± 23.9	3.9 ± 15.7	14.5 ± 20.1	1.8 ± 2.9
Quantile_10	59.6 ± 27.3	5.1 ± 25.7	14.1 ± 21.5	2.2 ± 4.1
Mean	75.7 ± 23.5	2.5 ± 9.4	16.6 ± 21.1	1.1 ± 2.7
FDCs	69.2 ± 24.7	4.1 ± 18.2	13.9 ± 20.2	2.4 ± 3.8
MLFD_Q10	41.1 ± 27.8	9.2 ± 28.0	10.6 ± 18.7	13.1 ± 15.5
MHFD_Q90	57.1 ± 26.8	4.2 ± 18.2	11.5 ± 18.6	11.6 ± 13.9
LFD_Q10	59.1 ± 28.0	5.6 ± 28.0	11.5 ± 20.1	3.6 ± 5.3
HFD_Q90	70.1 ± 24.2	3.5 ± 14.0	14.8 ± 19.8	3.9 ± 5.2

Note. The statistics above represent median ± standard deviations.

Figure 8 further summarizes the contributions of climate and vegetation change within each climate region. Similarly, precipitation is identified as the primary driver of global changes in streamflow signatures, although its relative contribution varies across climatic zones and types of streamflow signatures. In equatorial regions, all streamflow signatures are predominantly controlled by precipitation, with median contributions (hereafter referred to as "median") exceeding 60% for each streamflow signature, while the influences of temperature and LAI are relatively small. In contrast, the polar zone exhibits a stronger temperature influence, especially on low-flow signatures (61.4%–73.6%). Across all eight streamflow signatures in this zone, contributions of temperature range from 8.2% to 73.6%, followed by contributions of precipitation (17.1%–69.1%). In snow-covered regions, while precipitation remains the dominant driver (23.8%–80.9%), temperature contributions (3.9%–46.3%) are

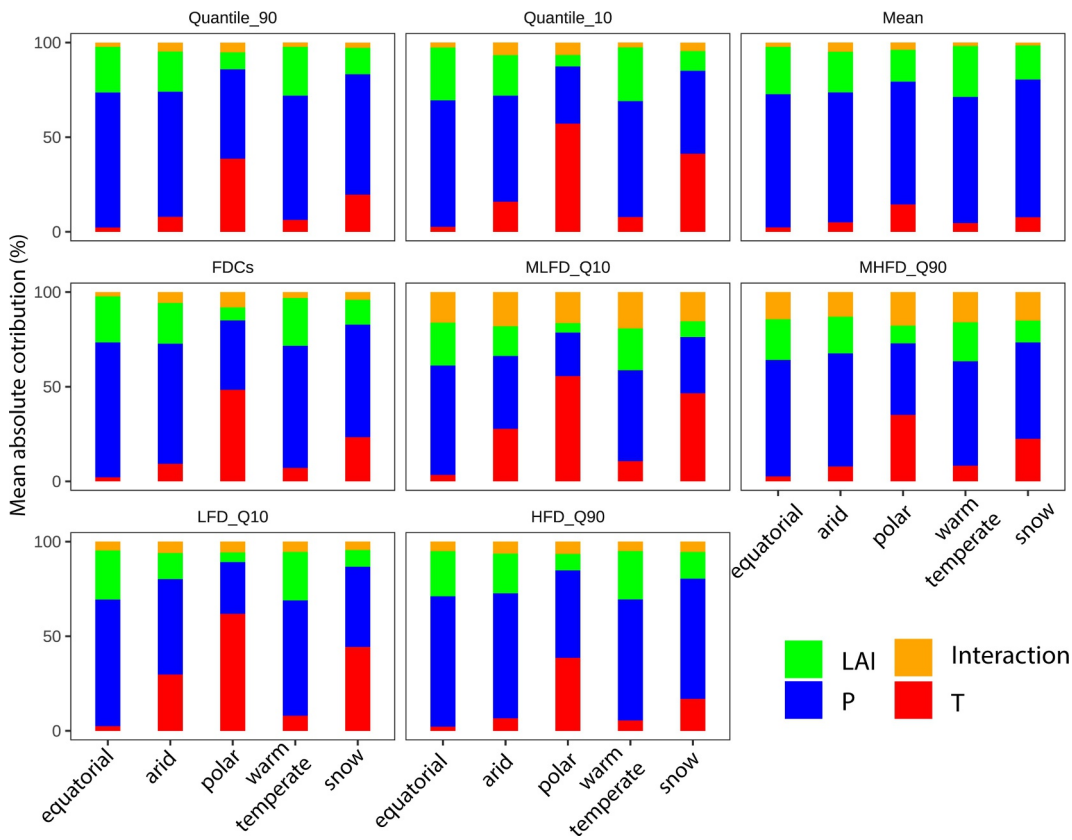


Figure 8. Absolute contributions of various factors to streamflow signatures across five climate zones.

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significant, ranking second only to those observed in the polar zone. In arid and warm temperate zones, the contribution patterns of precipitation and temperature are similar to those in the equatorial zone, though temperature contributes more than it in the equatorial zone.

Comparing the result with other existing studies, runoff in 83% of the land grid cells is most sensitive to precipitation changes (Berghuijs et al., 2017). In certain basins of snow regions, changes in runoff are predominantly driven by variations in precipitation (Rasouli et al., 2022), which is similar to the contribution of precipitation and temperature on mean streamflow in Figure 8. Increased temperature contributes to a 12% runoff decrease in three river source regions of China, while increased LAI causes a 33% of runoff decrease (Deng et al., 2023). The above ratio between the contribution of LAI and temperature of around 2.75 agrees well with our findings for snow regions (Figure 8, top right panel), validating the reliability of our attribution results. Since few studies have focused on the effects of climate and vegetation change on streamflow signatures, this study fills a critical gap by providing a comprehensive understanding of the contributions of climate and vegetation changes to streamflow signatures.

4. Discussions

4.1. Physical Mechanisms Driving the Divergent and Amplified Effects in Hydrological Propagation

Our results demonstrate that changes in meteorological and vegetation factors propagated nonlinearly to hydrological processes and were always amplified through total actual evapotranspiration-soil moisture-streamflow signature pathways and increasing temporal scales. These non-linear relationships are also reported in previous studies. In the Himalayan basin, temperature effects on runoff intensify non-linearly: 1°C, 2°C, and 3°C warming produce 7%, 19%, and 28% increases in total streamflow, evidencing a more-than-proportional amplification with rising temperature (Arora et al., 2008). Catchments in drier conditions are easier to attenuate climate seasonality to hydrological seasonality (Gnann et al., 2020). Compared to these studies, the novelty of our study is to explore the amplification effects throughout the entire hydrological propagation process across various temporal stages in different climate regions and multiple catchments, thereby providing more generalized insights.

Our modeling results exhibit the fundamental mechanisms driving the non-linear effects, as displayed in Section 3.3 for the complex process of vegetation dynamics and soil water transmission. On the one hand, these changes exhibit internal process complexity. For example, in the earlier moisture propagation chain, the increases in Et and Ei are partially offset by the decline in Es ((Y. Zhang et al., 2016), the first and last row of Figure 7), resulting in a dampened net change in actual evapotranspiration. This internal compensation results in a dampened net change in actual evapotranspiration. However, the reduced changes in evapotranspiration are later amplified in most cases through subsurface moisture pathways. On the other hand, these changes are time varying due to the deeply altered hydrological environments. For instance, vegetation does not immediately reduce its water consumption proportionately during drought spells. Instead, deep-rooted vegetation can sustain transpiration by actively extracting deeper soil moisture and groundwater (Fan et al., 2017). Consequently, even a marginal decrease in precipitation or a slight increase in vegetation greening can drastically deplete the subsurface storage that sustains baseflow. As vegetation intercepts and consumes the available water, the threshold for generating continuous streamflow is elevated, thereby translating a minor environmental anomaly into an amplified change in hydrological process. Consequently, low flow is predominantly controlled by groundwater discharge and slow subsurface flow routing. Enhanced depletions of these components driven by vegetation greening often persist, thereby resulting in more sensitive streamflow duration signatures compared to streamflow magnitude signatures.

4.2. Methodological Uncertainties and the Robustness of the Attribution Framework

While the proposed Scheme 1 based on HBV-PML demonstrates robust performance and clear improvements, it is important to acknowledge the inherent uncertainties associated with any single-model framework. Like most conceptual hydrological modeling approaches, our simulations involve necessary simplifications regarding model inputs (Y. Zhou et al., 2021), structural representations of hydrological processes (Kempen et al., 2021) and parameter equifinality issues (Moges et al., 2021). While the current model captures the essential hydrological dynamics effectively, exploring a multi-model ensemble approach in future studies would be a valuable next step to address these common uncertainties and further enhance overall model robustness.

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To further address the uncertainties inherent in our model-dependent attribution framework, three independent baseline methods were employed to cross-validate the results of HBV-PML. These include a semi-empirical physical framework (an improved Budyko model derived from Choudhury et al., 1999; Li et al., 2013; Oudin et al., 2005), and two purely data-driven algorithms (a Multiple Linear Regression (MLR) model and a Random Forest (RF) model). To match the temporal scale of the streamflow signatures, daily forcing data were aggregated into targeted annual metrics prior to training. The Budyko framework, which focuses exclusively on mean annual streamflow, was calibrated independently for each catchment. In contrast, to ensure robust statistical learning across sufficient sample sizes, the MLR and RF models were trained regionally for each climate zone, with individual models constructed for each specific streamflow signature. Detailed model formulations and performances are provided in Text S2 and Figure S10 in Supporting Information S1. The three methods outperformed HBV-PML in several streamflow signatures in calibration sets. However, they degraded dramatically in validation sets. In summary, the process-based model, HBV-PML, possessing a larger range of model performance value, agrees well with other methods, especially in mean streamflow prediction. It also provides continuous hydrological process and derives all streamflow signatures from one model at the same time. Meanwhile, it performs more stably when comparing calibration and validation sets, compared to the other three models.

Furthermore, a global-scale sensitivity of streamflow signatures to climate and vegetation changes is also conducted to compare the results of HBV-PML with other methods (Figure S11 in Supporting Information S1). Using stratified analysis to obtain response trend lines for global median responses, we found that HBV-PML might underestimate the streamflow sensitivity to temperature changes. However, it also reveals that all methods agree well with the directional response of streamflow signatures to precipitation, temperature, and LAI.

In addition, the consistency of the portion of dominant factors between HBV-PML and the other three methods is further evaluated in Figure S23 in Supporting Information S1. A clear conclusion can be drawn that HBV-PML performs similarly on the portion of dominant factors in mean streamflow compared to the improved Budyko framework. As for other methods, all of them agree with HBV-PML that precipitation dominates changes in streamflow signatures in most catchments. However, the dominance of temperature and vegetation varies among different methods and climate regions.

Finally, the consistency of mean absolute contributions of various factors to streamflow signatures using different methods is presented in Figure S24 in Supporting Information S1. The results clearly identify precipitation as the dominant driver of changes across all streamflow signatures. Meanwhile, the contributions of temperature, LAI and climate-vegetation interactions remain highly complex, underscoring the need for careful quantification in future studies.

4.3. Complex Climate-Vegetation Feedback and Future Perspectives

The climate- and vegetation-driven responses of streamflow signatures across climate zones are broadly consistent with existing evidences, however, several limitations should be further acknowledged. The HBV-PML model used in this study represents a relatively loosely coupled framework linking vegetation, climate, and hydrology, where vegetation dynamics (LAI) are externally provided by observations or detrended time series. Because observed LAI inherently embeds the outcomes of CO_2 fertilization, our methodology cannot mathematically isolate the CO_2 fertilization effect from the overall LAI variations. Consequently, the streamflow changes attributed to LAI in our results inevitably reflect the effect of CO_2-driven structural changes. This design overlooks potential feedbacks of vegetation change on local and regional climate (Miralles et al., 2025; Zeng et al., 2018) and the model implicitly assumes that vegetation and climate changes act as independent drivers of hydrological variation. Such simplification may lead to conservative estimates of vegetation impacts, as vegetation–climate feedback could either amplify or counteract the direct climatic influences on streamflow. For instance, precipitation sensitivity to greening is 0.624% yr^{-1} and −0.618% yr^{-1} for drylands and wet regions, respectively (Zhang, Gentine, et al., 2022). Studies have also demonstrated that increased LAI has contributed to warming boreal zones by reducing surface albedo and cooled arid regions in the Southern Hemisphere by boosting surface evaporation (Forzieri et al., 2017). Furthermore, vegetation greening also intensifies moisture recycling over wet regions via stimulating both evapotranspiration and precipitation (Zeng et al., 2018). Conversely, deforestation disrupts large-scale atmospheric circulation and moisture transport through increasing sensible heat flux (Makarieva et al., 2022). As a result, explicitly disentangling the complex, interactive physiological and structural effects of CO_2 from other climate variables requires a fully coupled dynamic vegetation

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model setup, and is beyond the scope of this study. Consequently, future research would benefit from incorporating dynamic vegetation modules or adopting fully coupled land-atmosphere frameworks to explicitly represent these interactions and more accurately quantify hydrological amplifying or buffering effects under a changing climate.

5. Conclusions

This study aimed to investigate the propagating effects of LAI, precipitation, and temperature on hydrological processes, especially clarifying the propagation effects of climate and vegetation changes on hydrological processes across different climate regions. Constrained by multiple streamflow signatures, the vegetation-dynamic-incorporated model, HBV-PML, achieved a robust overall performance (median KGE = 0.65). Through sensitivity experiments and stratified analysis, we disentangled the propagation effects and contributions of climate and vegetation changes to streamflow signatures among climate zones. Results show that the sensitivity of streamflow signatures to each driver varies by climate zone, with precipitation being the dominant driver in most conditions. Temperature shows a strong control on low-flow dynamics in polar regions. From the perspective of mechanisms, an increase in LAI enhances interception and transpiration, reduces soil moisture, and leads to lower streamflow magnitudes and more persistent low-flow conditions. Precipitation shows opposite effects on streamflow signatures, increasing water availability and streamflow signatures. Temperature affects hydrology primarily through changes in rain–snow partitioning and evapotranspiration, reducing high flows and amplifying low flows, thereby equalizing the streamflow extremes. Furthermore, changes in climate and vegetation show clear amplification along both increasing temporal scales and hydrological pathways. The quantified contribution analysis shows that precipitation dominates in 73.6 ± 10.5% of catchments among eight streamflow signatures, particularly in equatorial regions where median contributions exceed 60%. However, in polar climates, temperature plays a leading role in shaping low-flow signatures. These findings emphasize the necessity of using large samples to analyze quantified propagation mechanisms among climate regions under the changes in climate and vegetation. The cross-model uncertainty analysis supports the overall conclusion, despite inevitable statistical differences among different methods. Although we have investigated large-sample hydrological impacts from climate and vegetation change, substantial data gaps remain in data-sparse regions, especially across Africa and Asia. The complex impacts in these areas deserve more in-depth study.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The model input data used in this study are publicly available, including MSWEP V2.8 (https://www.gloh2o.org/mswep/), MSWX (https://www.gloh2o.org/mswx/), GLASS V6 (https://glass.hku.hk/overview.html). The streamflow data of the 2,252 catchments used in this study were sourced from multiple Catchment Attributes and Meteorology for Large Sample Studies data sets, and their full details are provided in the Supporting Information. Furthermore, the core code, the input samples, the complete information and time series of streamflow signatures for these 2,252 catchments are archived at zenodo (Huang & Zhang, 2026) and github (https://github.com/Kaye-HQ/PropagationEffects_Clim_Veg). Data processing, modeling, and most visualizations were conducted using *R* version 4.3.0 (https://cran.r-project.org/) within the RStudio environment (https://posit.co/download/rstudio-desktop/). The HBV-PML model can be found as an R package in github (https://github.com/Kaye-HQ/hbv.pml). The implementation of the PML-V2 and HBV models was based on the original open-source codes/software provided by the model developers, available at the websites of https://github.com/gee-hydro/gee_PML and https://www.geo.uzh.ch/en/units/h2k/Services/HBV-Model/HBV-Download.html, respectively.

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